12TH BIOLOGY · CHAPTER 24 · SOLVED EXERCISE

Evolution

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

SECTION 1: MULTIPLE CHOICE QUESTIONS

1. Who developed a theory of natural selection essentially identical to Darwin's?

(a) Hardy-Weinberg
(b) Alfred Wallace
(c) Malthus
(d) Lyell

Answer: (b) Alfred Wallace — Wallace's proposed mechanism was very similar to Darwin's; both were presented to the Linnaean Society in 1858.

2. Which statement is incorrect?

(a) Homologous organs are functionally different but structurally alike
(b) Analogous organs are functionally different but structurally alike
(c) Examples of analogous structures are arms of a man, forelimb of a cat, flipper of a whale
(d) Examples of homologous structures are wings of bats, birds and insects

Answer: Statements (b), (c) and (d) are all incorrect; only (a) is a correct statement. Analogous organs are alike in function but different in structure, so (b) is wrong. Arm, forelimb and flipper are homologous, so (c) is wrong. The insect wing is not homologous to the wings of birds and bats, so (d) is wrong.

Textbook error: the question has three incorrect statements, so no single option can be chosen. It was probably meant to ask "Which statement is correct?", for which the answer is (a).

3. Which of the following is an example of homologous organs?

(a) Wings of bird and insect
(b) Forelimbs of human and whale
(c) Fin of fish and wing of bird
(d) Eye of octopus and human

Answer: (b) Forelimbs of human and whale — Vertebrate forelimbs have a common arrangement of similar bones but different functions.

4. According to the theory of natural selection, variations arise due to;

(a) Mutations
(b) Use and disuse of organs
(c) Selective breeding
(d) Splitting of population

Answer: (a) Mutations — Genetic variations arise by random mutations.

5. Which condition is NOT required for Hardy-Weinberg equilibrium?

(a) Random mating
(b) No mutation
(c) Natural selection
(d) Large population size

Answer: (c) Natural selection — All genotypes must have an equal chance of survival and reproduction, i.e., there must be no natural selection.

6. Genetic drift is the change in gene frequency due to;

(a) Chance events (b) Natural selection
(c) Emigration of individuals (d) Immigration of individuals

Answer: (a) Chance events — Changes by chance, not fitness, are genetic drift.

7. Which of the following ideas is common to both Lamarckism and Darwinism?

(a) Mutations are the source of genetic variations
(b) Adaptations enable individuals to cope with environmental challenges
(c) Use and disuse of organs results in the development of acquired characters
(d) Acquired characters are inherited to next generations

Answer: (b) Adaptations enable individuals to cope with environmental challenges — Both theories explain evolution through adaptation to the environment; (a) belongs only to Darwinism, (c) and (d) only to Lamarckism.

8. If a massive fire kills 90% of a population and survivors remain by luck. Such a change in allelic frequency will be called;

(a) Natural Selection (b) Genetic Drift
(c) Gene Flow (d) Mutation

Answer: (b) Genetic Drift — The survivors remain by chance, not because they are better adapted.

9. Changes in gene frequency due to migration of individuals are called;

(a) Genetic drift (b) Mutation
(c) Gene flow (d) Genetic recombination

Answer: (c) Gene flow — Immigration and emigration change gene frequencies.

10. What may be the best evidence for the endosymbiotic evolution of eukaryotes?

(a) Presence of a nucleus in eukaryotes
(b) Circular DNA and 70S ribosomes in mitochondria
(c) Single-layered membranes around mitochondria
(d) Large volume of eukaryotic cells

Answer: (b) Circular DNA and 70S ribosomes in mitochondria — Mitochondria have their own circular DNA like a prokaryotic chromosome.

In the textbook, the options of MCQs 4 and 6 are labelled (a), (b), (d), (d); the third option is taken as (c). The 70S ribosomes of MCQ 10 are not mentioned in the chapter.

SECTION 2: SHORT QUESTIONS

1. What is the concept of evolution?

Evolution is the gradual change in the genetic makeup (allele frequencies) of a population over successive generations, which may lead to the formation of new species. In its earliest form, the concept had these basic speculations:

- The universe and organisms did not always exist in their present form.
- Today's organisms were not created at once.
- They are the result of innumerable changes from the lower to the higher form.

2. How does homology provide an evidence of divergent evolution?

Homologous structures are similar in structure but different in function. For example, the forelimbs of birds, bats, whales and horses have a common arrangement of similar bones, but are used for flight, swimming and running. The same basic structure shows that these organisms have a common ancestor; the different functions show that they evolved into different types. This is divergent evolution.

3. How analogous organs represent convergent evolution?

Analogous organs are similar in function but differ in structure, e.g., the wing of a bird and the wing of an insect. The different structure shows that the organisms evolved from different ancestors; the same function shows that they adapted in the same way in a common habitat (to fly, a broad, flat gliding surface is needed). This is convergent evolution.

4. What is Hardy-Weinberg theorem?

The **Hardy-Weinberg theorem** was presented in 1908 by the English mathematician Godfrey Hardy and the German physician Wilhelm Weinberg. It states that the frequencies of alleles and genotypes in a large, randomly mating population remain constant from generation to generation, if no evolutionary forces are acting.

5. What are the factors that change allele frequencies?

1. **Genetic drift (neutral selection):** change in gene frequencies by chance events, most obvious in small populations.
2. **Gene flow:** change in gene frequencies due to the migration of individuals into or out of a population.
3. **Mutation:** the origin of all new genes and a source of variations.

Natural selection also changes allele frequencies, because all genotypes do not have an equal chance of survival and reproduction.

6. Justify Lamarck as an early proponent of evolution.

Lamarck (1744–1829) published *Philosophie Zoologique* in 1809, fifty years before Darwin's "On the Origin of Species" (1859). He presented the idea of progression: organisms changed from the simplest to complex, and changes occur when organisms adapt to their environment. He also proposed a mechanism for this change (use and disuse of organs, and inheritance of acquired characters). Although his mechanism was later disproved, he was among the first to advocate evolutionary change.

7. Describe the drawbacks of Lamarckism.

After the mechanisms of inheritance were discovered, scientists disproved Lamarckism on the following bases:

1. Acquired characters have no genetic basis and therefore cannot be inherited by the next generations.
2. Organs are not modified by the wish or requirements of an organism.

8. Differentiate between;

• Lamarckism and Darwinism

Lamarckism	Darwinism
Proposed by Jean-Baptiste Lamarck in 1809 (Philosophie Zoologique).	Proposed by Charles Darwin in 1859 (On the Origin of Species).
Changes arise by use and disuse of organs (acquired characters).	Variations arise by random mutations.
Acquired characters are inherited.	Only inherited (genetic) variations are passed on; nature selects the adaptive traits.
Giraffe: the neck became long by stretching, and this was inherited.	Individuals with advantageous variations survive the struggle and reproduce.
Disproved; acquired characters have no genetic basis.	The most accepted theory; integrated with genetics as Neo-Darwinism.

• Homologous and analogous

Homologous structures	Analogous structures
Similar in structure but different in function.	Similar in function but different in structure.
Organisms have a common ancestor.	Organisms have different ancestors.
Reveal divergent evolution.	Indicate convergent evolution.
Example: forelimbs of birds, bats, whales and horses.	Example: wing of a bird and wing of an insect.

• Natural selection and neutral selection

Natural selection	Neutral selection (genetic drift)
Gene frequencies change because of fitness: individuals with adaptive traits survive and reproduce.	Gene frequencies change because of chance events, not fitness.
Acts in populations of any size.	Most obvious in small populations.
Makes the population better adapted.	Does not make the population better adapted; lowers genetic diversity.
Example: finches with beaks suited to the food of their island.	Example: a fire reduces 100 rabbits to 10; the survivors happen to be mostly brown.

SECTION 3: LONG QUESTIONS

1. Explain that evolution takes place due to variations in organisms and natural selection.

Evolution takes place due to **variations** in organisms and **natural selection**. Organisms with beneficial traits are selected to produce the next generations, while individuals with disadvantageous traits cannot produce next generations. As a result, the percentage of individuals with better traits increases in the population.

Variations

Genetic variations arise by random mutations, so seldom are any two individuals exactly alike. Variations may be advantageous, harmful or neutral, and inherited variations pass to the offspring. Mutations are the origin of all new genes; without variation there would be nothing to select.

Natural selection (selection pressure)

Organisms produce many more offspring than resources can support, so individuals struggle for survival. The environment selects: individuals with disadvantageous variations die or cannot reproduce, while those with advantageous variations (adaptive traits) reproduce. Adaptive traits are inherited and become more frequent; disadvantageous traits are eliminated. Over many generations, the accumulation of adaptations can make a population a separate species.

Examples

- **Galápagos finches:** variations in beak size and shape were selected according to the kind of food available on each island.
- **Antibiotic resistance:** a mutation in a bacterial gene lets some bacteria survive an antibiotic; the resistance gene, selected naturally, passes to the next generations.

2. Explain the theory of natural selection as proposed by Darwin.

Charles Darwin (1809–1882) was the founder of the modern theory of evolution. In 1831 he sailed as a naturalist on HMS Beagle on a 5-year voyage around the world. In 1859 he published “On the Origin of Species”, in which he proposed that evolution takes place due to variations in organisms through natural selection.

Darwin’s observations

- **Diversity and fossils:** In tropical rainforests and other new habitats he saw a great diversity of plants and animals. He found rocks containing fossils, which suggested that continents and oceans had changed dramatically over time.
- **Extinct mammals:** He dug up fossils of gigantic extinct mammals; organisms looked very different in the past. This suggested that living things change over time.
- **Galápagos Islands (16 small islands):** Plants and animals on different islands were very different, e.g., giant tortoises had saddle-shaped shells on one island and dome-shaped shells on another. Many of them resembled the plants and animals of South America.

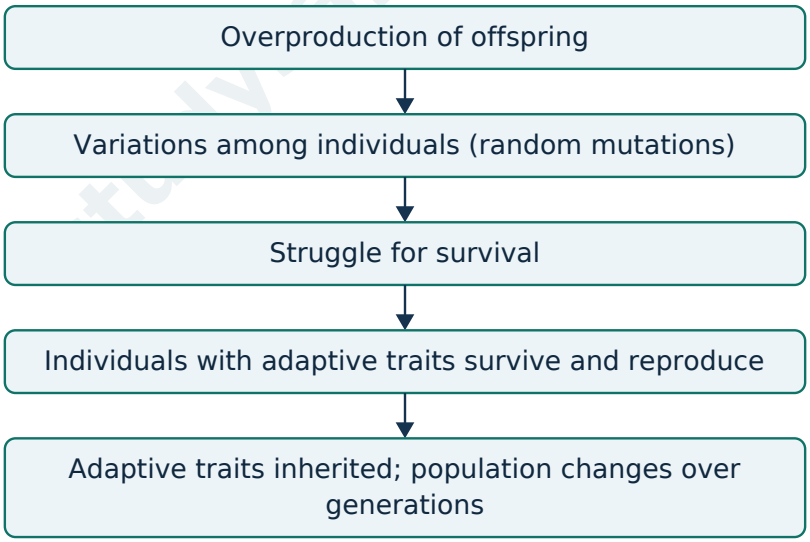
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- **Finches:** Different species of finches had different beaks. The sizes and shapes of the beaks were adaptations for feeding on completely different kinds of food on different islands.

His explanation: The origin of new species and adaptation are closely related processes. A new species arises from an ancestral form by the gradual accumulation of adaptations to a different environment, separated from the original habitat. Over many generations the two populations become dissimilar enough to be separate species.

Essence of the theory

1. **Overproduction:** All organisms have a far greater reproductive potential than is ever realized. For example, a female sea star releases about 1 million eggs each season; if all developed into adults, sea stars would fill the oceans in a few generations.
2. **Variations:** Genetic variations arise by random mutations. Seldom are any two individuals exactly alike. Variations may be advantageous, harmful or neutral. Inherited variations can be passed on to offspring.
3. **Struggle for survival:** Many more offspring are produced than resources can support, so individuals struggle for survival. Individuals with disadvantageous variations die or cannot reproduce; those with advantageous variations reproduce successfully. Such variations are called **adaptive traits**.
4. **Inheritance of adaptive traits:** Adaptive traits are inherited by the next generations. Disadvantageous traits become less frequent in the population and are eventually eliminated.



Natural selection as proposed by Darwin

3. Explain how biogeography provides evidence for evolution.

Biogeography is the study of the geographic distribution of plants and animals. It shows that life-forms in different parts of the world have distinctive evolutionary histories.

- **Similar groups separated by vast barriers:** Obvious similarities suggest a common ancestry, while obvious differences result from millions of years of

independent evolution. Example: the African lion is very similar to the mountain lion of America.

- **Different organisms in similar environments:** Organisms separated by geographical barriers are often very different in spite of similar environments. Example: many animals of Australia and Tasmania are very different from animals in any other part of the world; many species of kangaroos are the major herbivores there.
- **Islands:** Islands often have relatively few but unique species, due to island colonization and subsequent evolutionary events.

Darwin's own observations are of this kind: the plants and animals of the Galápagos Islands resembled those of nearby South America, yet differed from island to island.

4. Describe the evidences of evolution that come from comparative anatomy.

Comparative anatomists study the structure of fossils as well as living organisms, looking for similarities that indicate evolutionary relationships. This study reveals two types of evolution, and vestigial organs.

Divergent evolution: Body parts that are similar in structure but different in function are called **homologous structures**, and the phenomenon is **homology**. Homologous structures reveal divergent evolution: the organisms have a common ancestor, evolved into different types and used their structures for different functions. Example: vertebrate forelimbs have a common arrangement of similar bones, although their functions vary: flight (birds, bats), swimming (whales, dolphins), running (horses), climbing (arboreal lizards) or swinging from tree branches.

Convergent evolution: Organs that are similar in function but differ in structure are called **analogous organs**. They indicate convergent evolution, in which different types of organisms have evolved from different ancestors in a common habitat. Example: the wing of a bird and the wing of an insect are both adaptations for flight, but they are not homologous. The similarity only reflects that, to fly, an animal needs a broad, flat gliding surface.

Vestigial organs: Organisms often retain structures that have lost their usefulness. These poorly developed structures are called vestigial structures. Example: boa constrictors and pythons have minute remnants of hindlimb (pelvic) bones left over from their reptilian ancestors. Such remnants of once-useful structures are clear indications of change, i.e., evolution.

Vestigial structures in humans (about 90): the vermiform appendix (vestige of the cecum), the coccyx or tailbone (remnant of the tail), the wisdom teeth (vestigial third molars) and the ear muscles.

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5. Differentiate between convergent and divergent evolution on the basis of inheritance of the homologous and analogous structures.

Divergent evolution	Convergent evolution
Shown by homologous structures.	Shown by analogous structures.
Structures are similar in structure but different in function.	Structures are similar in function but different in structure.
Organisms have a common ancestor (structure inherited from it).	Organisms have evolved from different ancestors (structure not inherited from a common ancestor).
Organisms evolve into different types and use the structures for different functions.	Different organisms in a common habitat develop structures for the same function.
Example: forelimbs of birds, bats, whales, horses.	Example: wing of a bird and wing of an insect.

So homologous structures are inherited from a common ancestor and then modified for different functions (divergence), whereas analogous structures are not inherited from a common ancestor; they arise separately in unrelated organisms for the same function (convergence).

6. Describe the evidences of evolution that come from molecular biology.

Just as animals have homologous structures, they also have **homologous biochemicals**.

- Structure and function are ultimately based on the genetic blueprint (DNA) found in all living animals.
- Related animals have similar DNA, derived from their common ancestor.
- DNA carries the codes for proteins, so related animals also have similar proteins.
- Scientists extract and analyse the DNA and proteins of different organisms. By looking for dissimilarities in related proteins and DNA, and by assuming relatively constant mutation rates, they estimate the time elapsed since evolution from a common ancestral molecule.

7. Outline the steps of the evolution of the giraffe, as illustrated in Lamarckism.

1. The ancestors of the giraffe did not have long necks. They ate grass or the leaves of small plants.
2. The environment changed, and small plants were no longer available.
3. Some individuals began stretching their necks to reach the leaves of tall trees.
4. By constant stretching, their necks became strong and long during their lifetime: an acquired character.
5. When they reproduced, this acquired character was inherited; the young were born with slightly longer necks.

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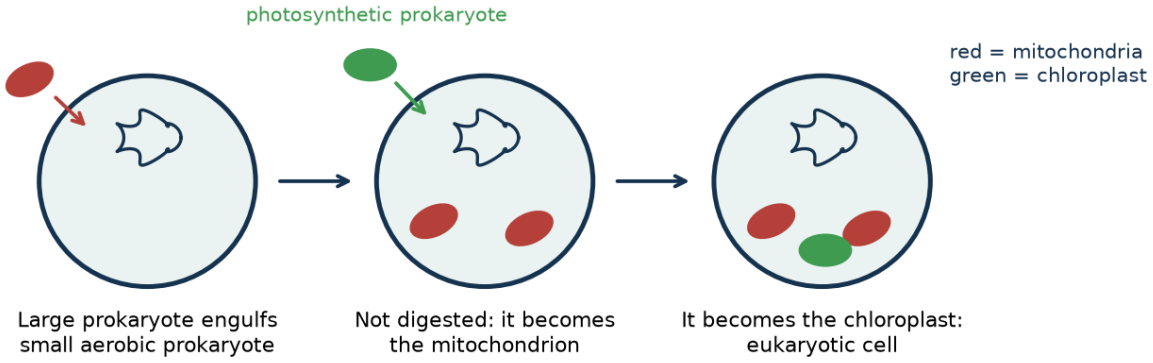
6. These individuals continued the same practice, and their necks became even stronger and longer.
7. After the character was inherited over many generations, the generation with long necks evolved.

8. Describe the endosymbiont theory about the evolution of eukaryotes from prokaryotes.

The **endosymbiotic theory**, proposed by **Lynn Margulis**, is the most acceptable theory for the evolution of eukaryotes from prokaryotes (about 2 billion years ago).

1. Large prokaryotes engulfed small prokaryotes.
2. The small prokaryotes were not digested. They lived within the large cells and evolved into organelles. Both cells benefited (symbiotic relationship).
3. Some small prokaryotes could carry out aerobic respiration. They supplied energy to themselves and to the large cell, and became the **mitochondria**.
4. Other small cells could use sunlight to make food, which they shared with the large cell. They became the **chloroplasts**.
5. Flagella may have been derived through the ingestion of prokaryotes similar to spirochetes.

The eukaryotic cells so formed were powerful and efficient. This enabled them to evolve new characteristics such as multicellularity, cell specialization and large size, which led to the diversity of animals, plants and fungi.



Endosymbiotic origin of mitochondria and chloroplasts

Evidence for the theory

- Mitochondria (and chloroplasts) are of the same size as prokaryotic cells.
- They have their own DNA, organized in a circular chromosome like a prokaryotic chromosome.
- They reproduce by binary fission like prokaryotes; eukaryotic cells reproduce by mitosis.
- Their membrane composition is more similar to prokaryotic membranes than to eukaryotic membranes.

9. Describe the theory of inheritance of acquired characters, as proposed by Lamarck.

The French naturalist **Jean-Baptiste Lamarck** (1744–1829) published the book *Philosophie Zoologique* (Zoological Philosophy) in 1809. In it he presented the idea of progression (change) known as **Lamarckism**: organisms changed from the simplest to complex, and changes occur in organisms when they adapt to the environment.

1. **Use and disuse of organs:** When an organism faces changes in its environment, it begins to use some organs more and disuses others. Organs used more frequently develop and become strong; organs not properly used deteriorate or become weak. Such changes, due to long-term use or disuse, are **acquired characters**.
2. **Inheritance of acquired characters:** The characters acquired during the lifetime of an organism are passed on to its offspring. Inherited from generation to generation, they ultimately modify the organism to cope better with the changes in the environment.

Example: The ancestors of the giraffe had short necks. When small plants were no longer available, they stretched their necks to reach the leaves of tall trees. The neck became long (acquired character), and this was inherited from generation to generation until long-necked giraffes evolved (steps in Q.7).

Drawbacks

After the mechanisms of inheritance were discovered, scientists disproved Lamarckism on the following bases:

1. Acquired characters have no genetic basis and therefore cannot be inherited by the next generations.
2. Organs are not modified by the wish or requirements of an organism.

INQUISITIVE QUESTIONS

1. Interpret different homologous and analogous structures through observation in plants.

Applying the textbook definitions to plants:

Type	Plant structures	Interpretation
Homologous	Tendrils of pea, spines of cactus (*Opuntia*) and fleshy scales of onion	All are modified leaves: same origin and basic structure, but different functions (climbing, protection, storage). Divergent evolution.
Homologous	Thorns of *Bougainvillea* and tendrils of grapevine or *Cucurbita*	Both are modified stem branches (axillary buds) with different functions (protection, climbing). Divergent evolution.
Analogous	Tendrils of pea (leaf) and tendrils of grapevine (stem)	Same function (climbing) but different origin and structure. Convergent evolution.

Type	Plant structures	Interpretation
Analogous	Potato (stem tuber) and sweet potato (root tuber)	Same function (food storage) but different origin. Convergent evolution.
Analogous	Spines of cactus (leaf) and thorns of *Bougainvillea* (stem)	Same function (protection) but different origin. Convergent evolution.

The textbook gives only animal examples; these plant examples are from general botany.

2. Hypothesize whether Lamarck was criticized in his day for advocating the ideas of evolution or for the mechanism he proposed.

Hypothesis: In his own day Lamarck was criticized more for advocating the idea of evolutionary change than for his mechanism.

Reasoning: At that time most people believed in creationism, i.e., that each type of organism was created individually and never changed, so the very idea that organisms change was rejected. Moreover, no evidence was available then that could explain the mechanism of evolution, so his mechanism could neither be tested nor disproved. Only later, when the mechanisms of inheritance were discovered, was the mechanism itself rejected, because acquired characters have no genetic basis.

3. Justify, on the grounds that both Wallace's and Darwin's papers were published in the Journal of the proceedings of the Linnaean Society, why the theory was attributed to Darwin.

- **Wallace himself agreed:** when both theories were presented to the Linnaean Society in 1858, Wallace agreed to attribute the theory to Darwin.
- **Darwin's evidence:** Darwin had collected observations for many years, beginning with his 5-year voyage on HMS Beagle (1831): fossils, the tortoises and finches of the Galápagos Islands, and the distribution of species.
- **Darwin's book:** in 1859 he published "On the Origin of Species", which explained the mechanism of natural selection in full, with evidence. The theory became known through this book.
- **Priority:** Darwin had been developing the idea long before Wallace's letter reached him.

The first three points are from the textbook; the fourth is general knowledge.

4. Describe and analyse examples of technology that have extended or modified the scientific understanding of evolution (e.g., the contribution of radiometric dating to the palaeontological analysis of fossils).

Technology	What it does	Contribution to understanding evolution
Radiometric dating	Measures the decay of radioactive isotopes in rocks and fossils	Gives the actual age of fossils, so they can be arranged in time order; shows that life is billions of years old (prokaryotes about 3.5 billion years).

Technology	What it does	Contribution to understanding evolution
DNA and protein sequencing	Reads the order of bases in DNA and of amino acids in proteins	Related organisms have similar DNA and proteins; the differences, with constant mutation rates, estimate the time since a common ancestor.
Electron microscopy and cell biology	Shows the fine structure of cells and organelles	Showed that mitochondria and chloroplasts have their own circular DNA and divide by binary fission: support for the endosymbiotic theory.
Genetics	Explains the mechanism of inheritance	Disproved Lamarckism and joined with Darwinism to form Neo-Darwinism (evolution as change in allele frequencies).
Computers and bioinformatics	Compare very large sets of DNA data	Allow evolutionary relationships among many species to be worked out.

Radiometric dating, electron microscopy and bioinformatics are not described in this chapter; these rows are general knowledge.

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